FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – II
PAPER –1
TEST DATE: 10-12-2023

ANSWERS, HINTS & SOLUTIONS

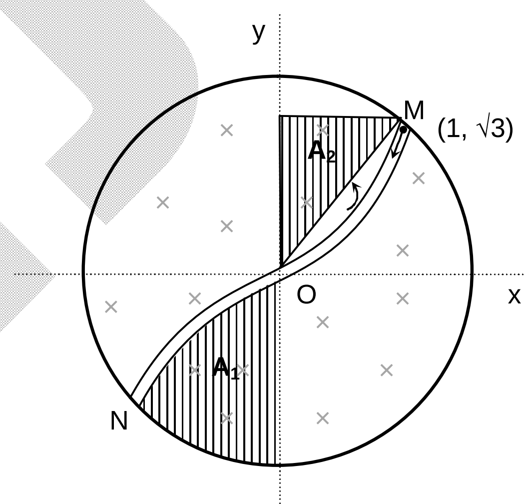
Physics

PART – I

SECTION – A

1. A, B, C, D

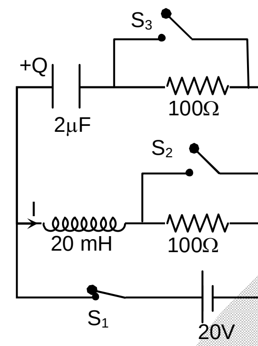
Sol.



$$y = \sqrt{3}x^2$$
$$dy = 2\sqrt{3}x\,dx$$
$$A_1 = \int x\,dy$$
$$A_1 = \int_0^1 2\sqrt{3}x^2 dx$$
$$A_1 = \frac{2\sqrt{3}}{3}$$
$$A_2 = \frac{1}{2}\times 1\times\sqrt{3} = \frac{\sqrt{3}}{2}$$
$$\phi = B(A_1 - A_2)$$
$$= B\left(\frac{2\sqrt{3}}{3} - \frac{\sqrt{3}}{2}\right) = \left(\frac{4\sqrt{3}-3\sqrt{3}}{6}\right)B = \frac{\sqrt{3}}{6}B$$
$$\varepsilon_{ind} = -\frac{d\phi}{dt} = \frac{\sqrt{3}}{6}\times 30 = -5\sqrt{3}$$

2. A, C

Sol. $Q = CV = 2 \times 10 \times 10^{-6} = 40\ \mu C$



$$U_C = \frac{1}{2}\frac{Q^2}{C} \Rightarrow 400\ \mu C$$

$$I = \frac{20}{100} = \frac{1}{5}A$$

$$U_L = \frac{1}{2}Li^2 = \frac{1}{2}\times 20\times 10^{-3}\times\frac{1}{25} = 400\ \mu J$$

$U = U_L + U_C = 800\ \mu C$

$$\frac{1}{2}\frac{Q_{max}^2}{C} = 800\mu C$$

$$Q_{max} = 40\sqrt{2}\ \mu C$$

$Q = Q_{max}\sin(\text{cot} + \phi)$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20\times 10^{-3}\times 2\times 10^{-6}}} = \frac{10^4}{2}t$$

$$Q = Q_0 \sin\left(\frac{10^4}{2}t + \phi\right) = 40\sqrt{2}\sin\left(\frac{10^4 t}{2} + \frac{3\pi}{4}\right)$$

$$\frac{1}{2}Li_{max}^2 = 800\times 10^{-6}$$

$$\frac{1}{2}\times 20\times 10^{-3} I_{max}^2 = 800\times 10^{-6}$$

$$L_{max} = \frac{2\sqrt{2}}{10} = \frac{\sqrt{2}}{5}A$$

at $t = 0$ and $t = \pi \times 10^{-4}$ sec

Energy stored in magnetic field = Energy stored in electric field

3. A, B, D

Sol. $\lambda_m T = b$

$\log_e \lambda = -\log_e T + \log_e b$, Slope = –1

$E = \sigma T^4$

$\log_e E = \log_e \sigma + 4\log_e T$, Slope = 4

$\log_e b = x$, $b = e^x$

4. B

Sol. $PV = 2RT$

$(A - BV^2)V = 2RT$

$AV - BV^3 = 2RT$

$$T = \frac{AV}{2R} - \frac{BV^3}{2R}$$

When T is maximum

$$\frac{dT}{dV} = 0, \quad \frac{A}{2R} - \frac{3BV^2}{2R} = 0$$

$3BV^2 = A$

$$V^2 = \frac{A}{3B} \Rightarrow V = \sqrt{\frac{A}{3B}}$$

5. C

Sol. $V_1 - \int_0^D E dx = V_2$

$QA\varepsilon_0 \left(x + \frac{x^3}{3B} \right) = \Delta V$

$C = \frac{Q}{\Delta V} = \frac{3B}{A\varepsilon_0 D(3B + D^2)}$

6. D

Sol. $R = \frac{24}{8} = 3$

$4.8 = \frac{24}{\sqrt{(3)^2 + \omega^2 L^2}} \Rightarrow L = 0.08$

$P = V_{rms} i_{rms} \cos \phi = 24$ watt

7. C

Sol. $P_0 V_0 = nRT_M$

$2P_0 V_0 = nRT_N$

$\Delta Q_{MN} = nC_P (T_M - T_N) = nC_P \frac{P_0 V_0}{nR} = 200\,\mathbf{J}$

$9P_0 V_0 = nRT_0$

$(\Delta U)_{OM} = nC_V (T_0 - T_M) = 900\,J$

$\frac{C_P}{C_V} = \gamma = \frac{16}{9}$

8. A

Sol. $x_L = \omega L, \qquad x_C = \frac{1}{\omega C}$

$\tan\phi = \frac{x_L - x_C}{R}$

9. D

Sol. (P) $E = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k} = \frac{1}{x^2}\hat{i} + \frac{1}{y^2}\hat{j} + \frac{1}{z^2}\hat{k}$

(Q) i = 4 – 4t, at t = 0 constant = 0

$H = \int_0^1 i^2 R d\tau = \int_0^1 (4 - 4t)^2 6 dt = 32$

(R) i = 4A

(S) R = 5Ω

10. A

Sol. (P) $P_0V_0^{5/3} = 32P_0V_2^{5/3}$

$V_2 = \frac{V_0}{(32)^{3/5}} = \frac{V_0}{8}$

$V_1 = V_0 + \frac{7V_0}{8} \Rightarrow \frac{15V_0}{8}$

$P_1 = 32P_0$

$P_1V_1 = 32P_0 \times \frac{15V_0}{8} = 60P_0V_0$

(Q) Work done on gas, $\frac{P_RV_R - P_0V_0}{\gamma - 1} = \frac{32P_0\frac{V_0}{8} - P_0V_0}{\frac{5}{3} - 1}$

$\Rightarrow \frac{3P_0V_0 \times 3}{2} = \frac{9P_0V_0}{2}$

(R) For left chamber

$\frac{P_0V_0}{RT_0} = \frac{60P_0V_0}{RT_\perp}$

$T_1 = 60T_0$

$\Delta U = nC_V\Delta T = 1 \times \frac{3R}{2} \times 59T_0 \Rightarrow \frac{177}{2}RT_0$

$\Delta U = \frac{177}{2}P_0V_0$

(S) Work done by gas in left chamber $\Rightarrow \frac{9}{2}P_0V_0$

$\Delta Q = W + \Delta U = \frac{9}{2}P_0V_0 + \frac{177}{2}P_0V_0 = 93\ P_0V_0$

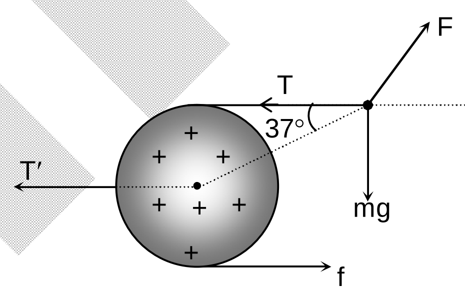
11. C

Sol. F sin 37° = mg

T = F cos 37°

f = T

T′ = 2T



SECTION – B

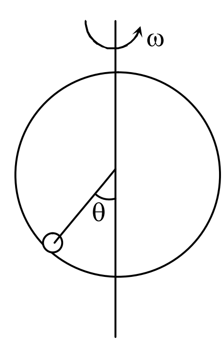
12. 3

Sol. $\tan\theta = \frac{x\omega^2}{g}$

$\frac{\sin\theta}{\cos\theta} = \frac{R\sin\theta\omega^2}{g}$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

$M = \frac{q\omega x^2}{2} = 3 \times 10^{-4}$



13. 4

Sol. When both current are flown simultaneously

$I = I_{DC} + I_{AC} \sin \omega t$

AC ammeter shows rms current

AC ammeter gives

$$I_{rms} = (I_{DC} + I_{AC} \sin\omega t)_{rms} = \sqrt{(I_{DC} + I_{AC} \sin\omega t)^2_{avg}}$$

$$= \sqrt{I^2_{DC} + (I_{AC})^2_{rms}} = \sqrt{6^2 + 8^2} = 10$$

14. 20

Sol. $mg - P_0A + PA = ma$

$mg - P_0A + P_0A - \frac{P_0A}{\ell}x = ma$

$$mg - \frac{P_0Ax}{\ell} = mV\frac{dV}{dx}$$

$$\int mgdx - \int \frac{P_0Ax}{\ell}dx = \frac{mv^2}{2}$$

$$mgx - \frac{P_0Ax^2}{2} = 0$$

$$10x - \frac{P_0Ax^2}{2\ell} = 0$$

15. 6

Sol. $$\frac{1}{2}mv_0^2 = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{d}$$

$$d = \frac{2Q^2}{4\pi\varepsilon_0 mv^2}$$

$$mv_0\frac{d\sqrt{3}}{2} = mv_1r$$

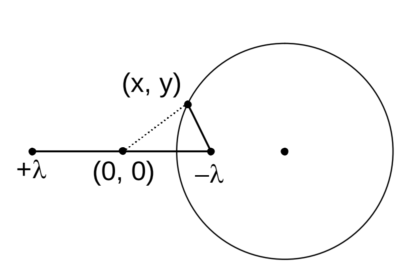
$$v_1 = \frac{3v_0d}{4r}$$

$$\frac{1}{2}mv_0^2 = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{r} + \frac{1}{2}mv_1^2$$

$$r = \frac{3d}{2}$$

16. 2

Sol. $$V = \frac{\lambda}{4\pi\varepsilon_0}\ell n\left[(a+x)^2 + y^2\right] - c - \frac{\lambda}{4\pi\varepsilon_0}\ell n[(a-x)^2 + y^2] + c$$



$(a+x)^2 + y^2 = 2(a-x)^2 + 2y^2$

$\Rightarrow (x - 3a)^2 + (y - 0)^2 = (2\sqrt{2}a)^2$

Centre of circle (3a, 0)

$R = 2\sqrt{2}a$

17. 70

Sol. emf $B\ell v$

$$i = \frac{B\ell v}{10 + (10 - 10x)}$$

$$-mv\frac{dv}{dx} = iB\ell = \frac{B^2\ell^2 v}{(20 - 10x)}$$

$v = 100\ \ell n2 \Rightarrow 70$ m/s

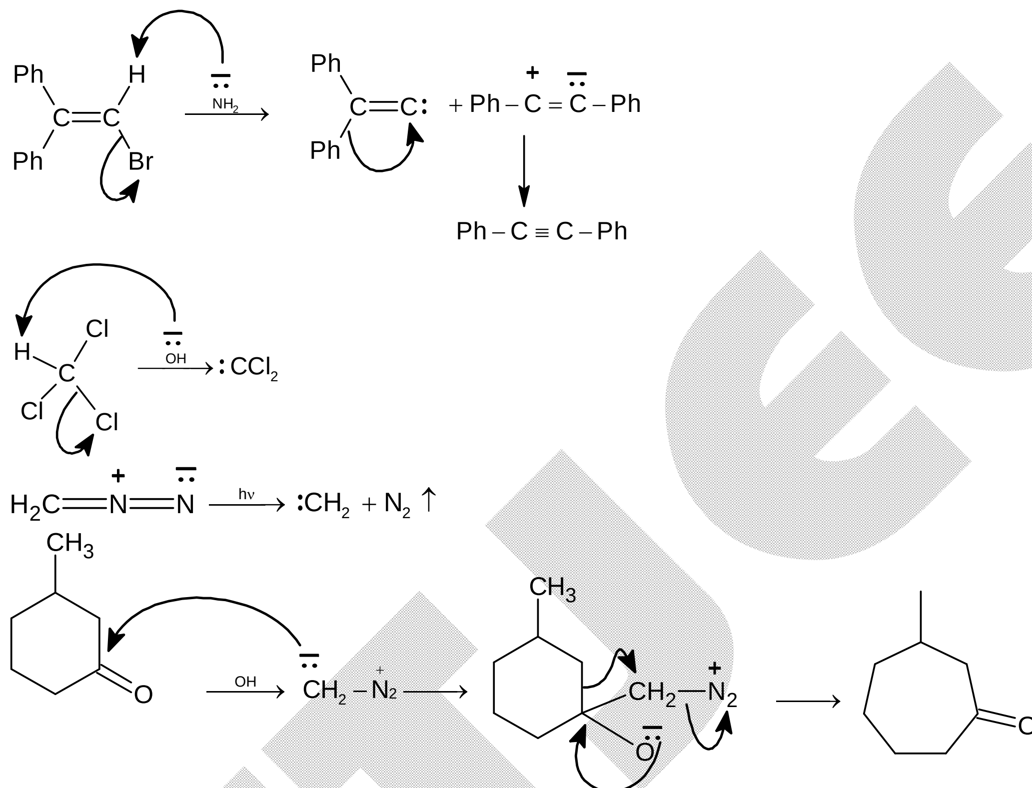
Chemistry

PART – II

SECTION – A

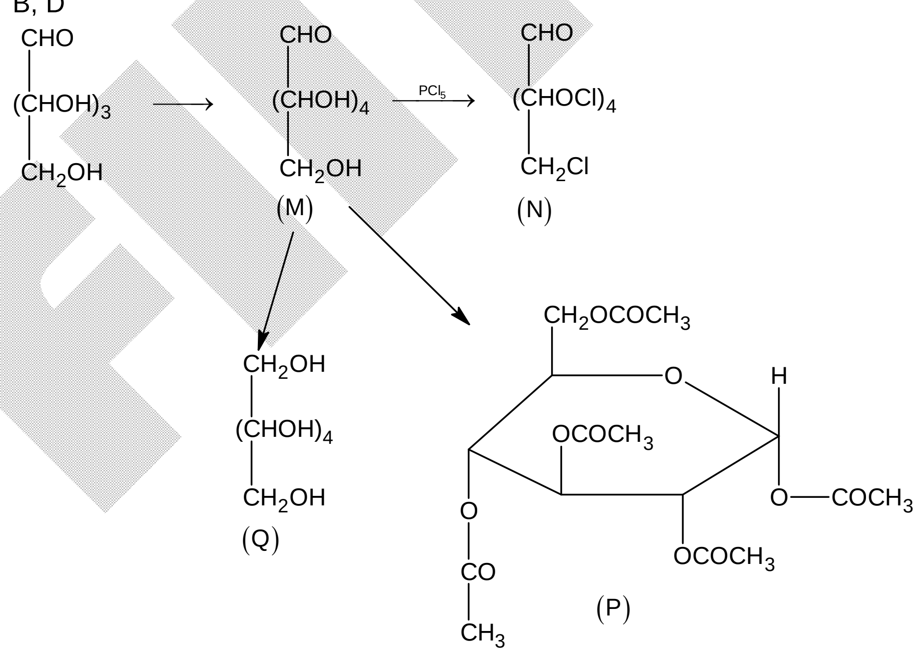
18. A, B, C

Sol.



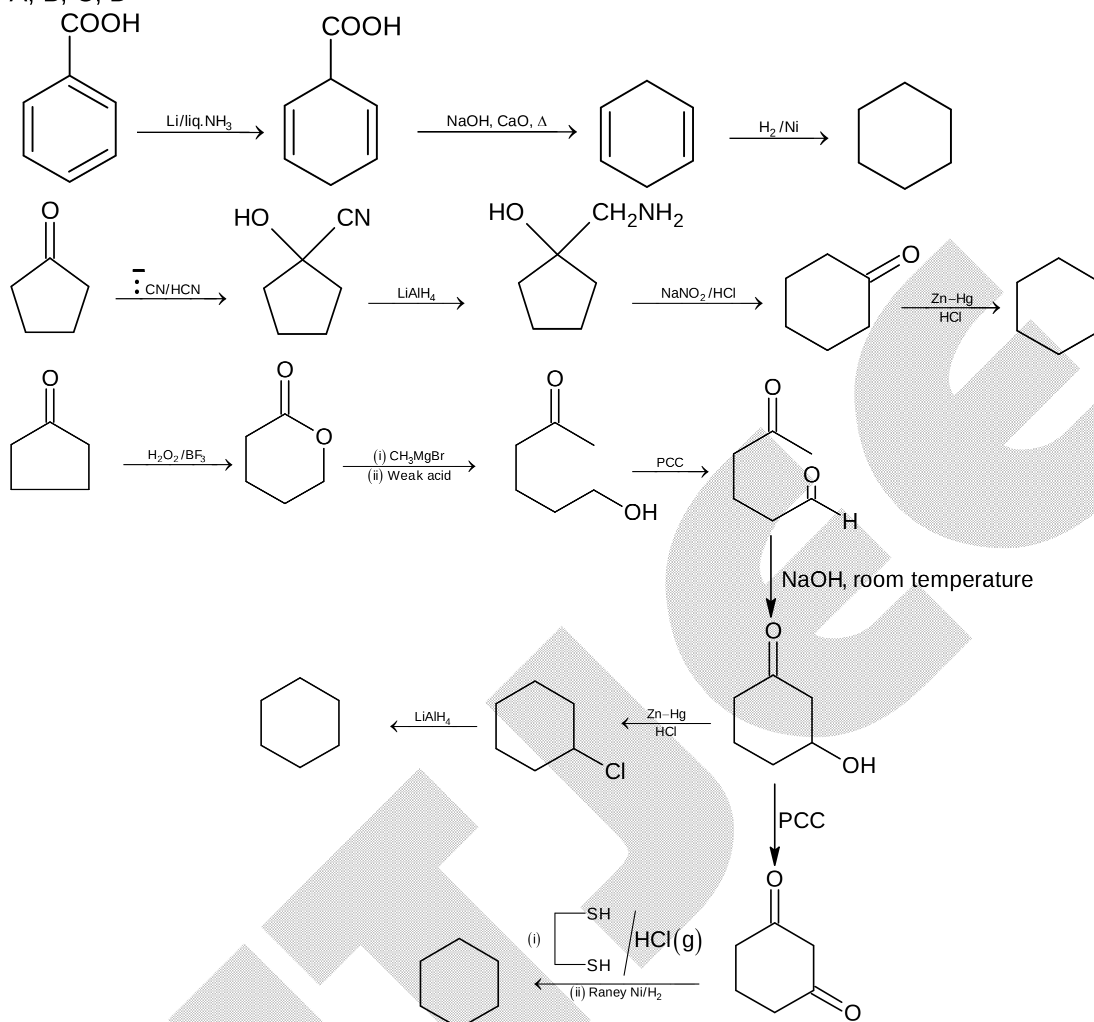
19. B, D

Sol.



20. A, B, C, D

Sol.



21. D

Sol. Starch is a polymer of $\alpha-D(+)$-glucose

Cellulose is a polymer of $\beta-D(+)$-glucose

Maltose is made up of 2 molecules of $\alpha-D(+)$-glucose

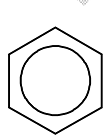
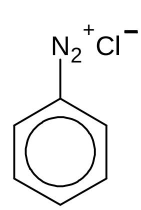
Sucrose is made up of $\alpha-D(+)$-glucose and $\beta-(D)(-)-$fructose

$\beta-$Lactose is made up of $\beta-D(+)$-galactose and $\beta-D(+)$-glucose

$\alpha-$Lactose is made up of $\beta-D(+)$-galactose and $\alpha-D(+)$-glucose

22. C

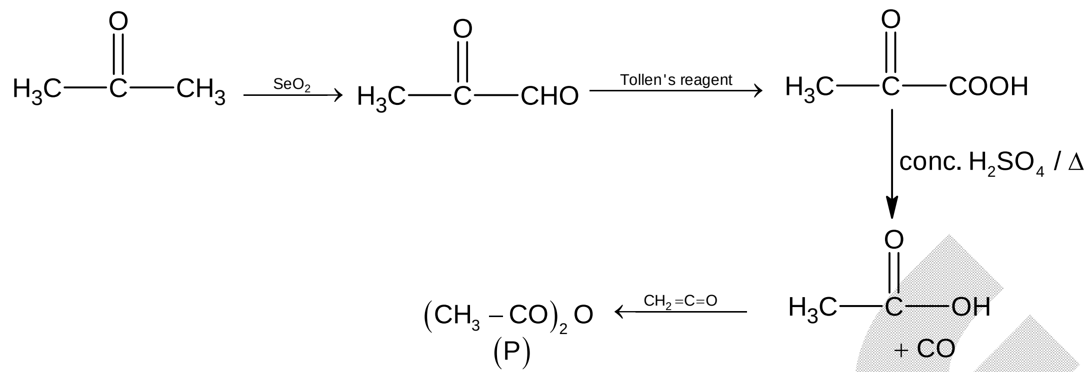
Sol.

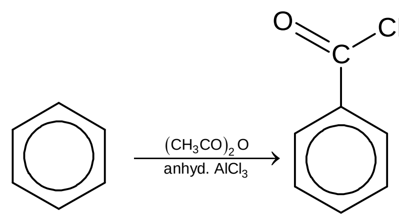
 , N_2H_4, 

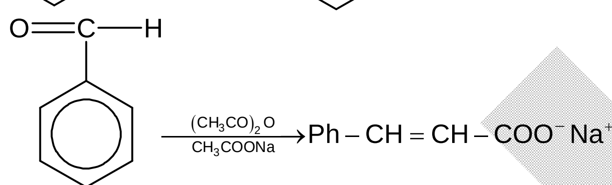
, HN_3 these do not give Lassaigne's test.

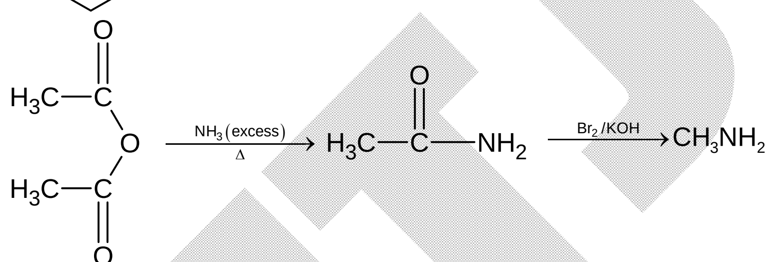
23. D

Sol.

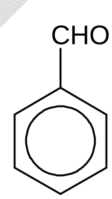


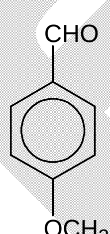


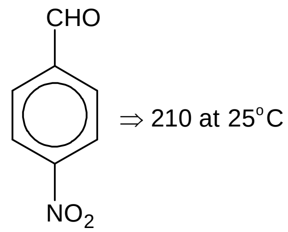




24. D

Sol. Rate of Cannizzaro reaction with respect to 

 $\Rightarrow$ 0.5 at 25^{o}C



Rate of S_N2 of RCl with Iodide ion

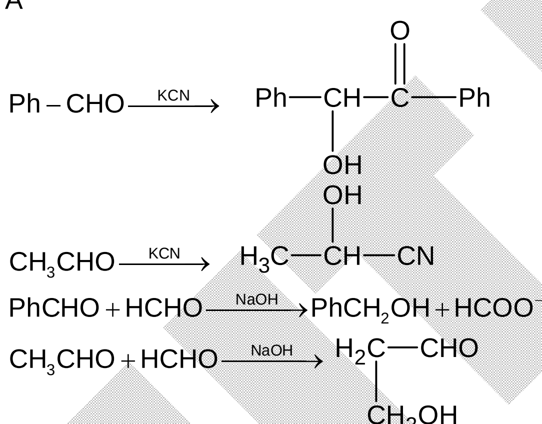
$CH_3Cl \Rightarrow 200, Ph-\overset{\overset{O}{||}}{C}-CH_2Cl \Rightarrow 10^5$

Pinacol – Pinacolone rearrangement takes place in hot and dil. H_2SO_4 solution.

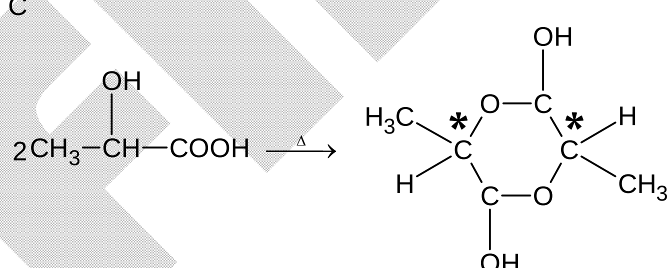
25. B
Sol. Facts

26. B
Sol. CH_3CH_2OH gives iodoform test
CH_3COOH gives off CO_2 from $NaHCO_3$ solution
PhCHO gives Tollen's test but not Fehling's test
PhOH give Br_2 water test result in yellow ppt.

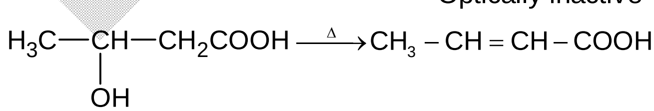
27. A
Sol.

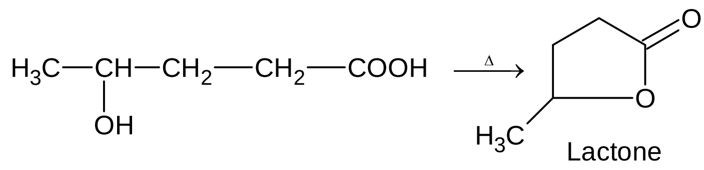


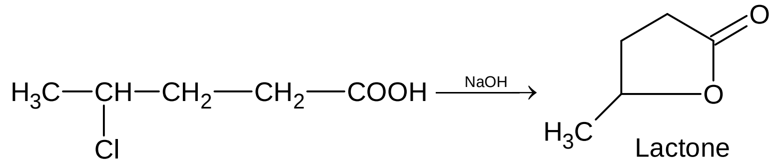
28. C
Sol.



Optically inactive



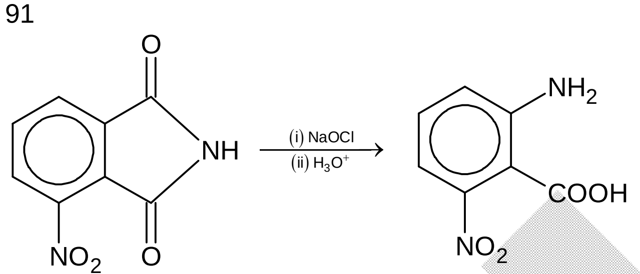




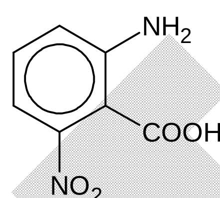
SECTION – B

29. 91

Sol.



Molecular mass of

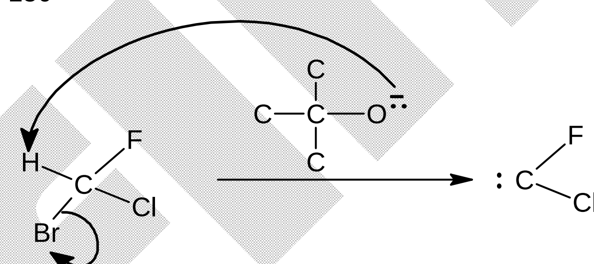


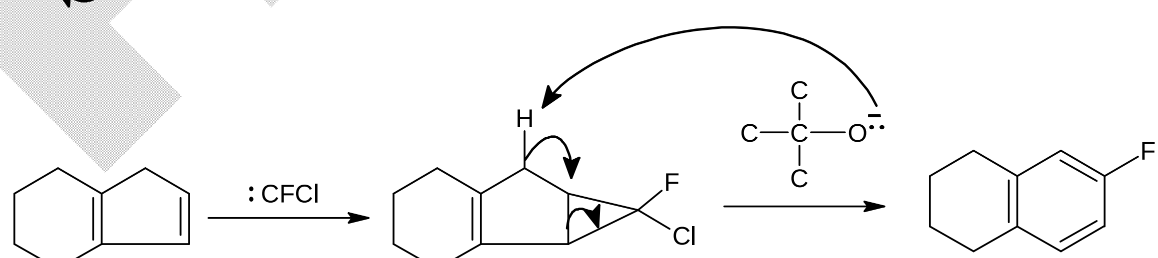
is 182

$\therefore\ W_{formed}$ is $182 \times \dfrac{50}{100} = 91\,\text{gm}$

30. 150

Sol.

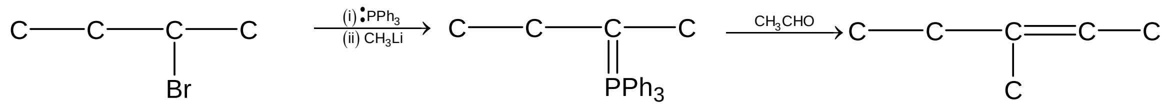




Mol. wt. = 150

31. 63

Sol.

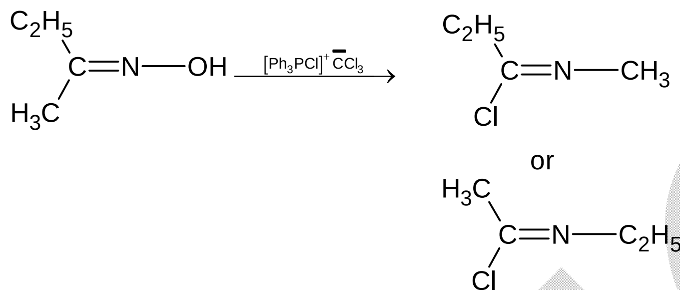


Mol. wt. $= 84$

$\therefore$ 75% yield $= 84 \times \dfrac{75}{100} = 63$ gm

32. 12

Sol. $Ph_3P + CCl_4 \longrightarrow [Ph_3PCl]^+ \overset{-}{C}Cl_3$

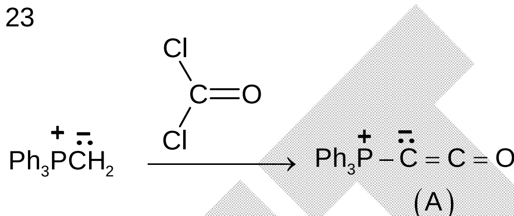


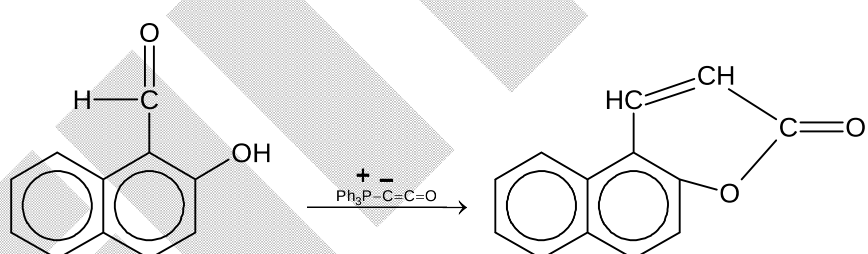
$\therefore x = 8, y = 4$

$\therefore x + y = 12$

33. 23

Sol.





$x = 20$

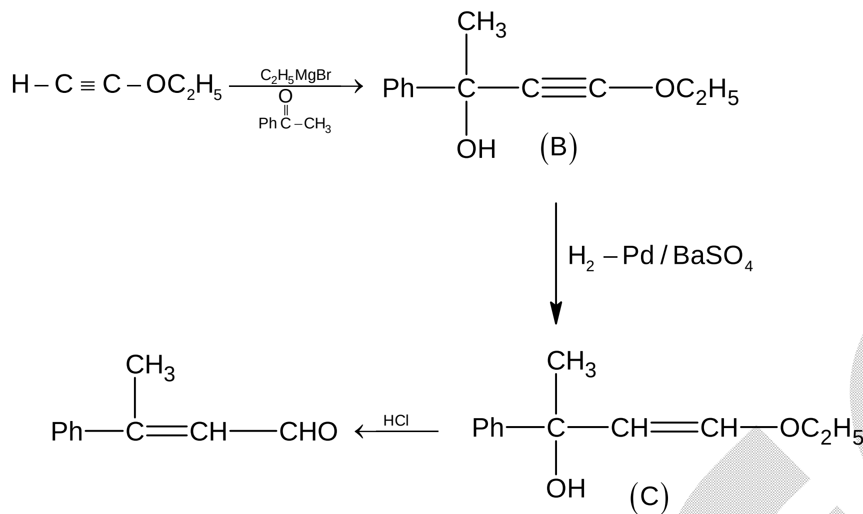
$y = 0$

$z = 3$

$\therefore x + y + z = 23$

34. 15

Sol.



$\therefore x = 2$

$y = 12$

$z = 1$

$\therefore x + y + z = 12 + 2 + 1 = 15$

Mathematics

PART – III

SECTION – A

35. A, B, C

Sol. $p = \dfrac{\left(1-\sin^2\alpha\sin^2\beta\right)^2\left(\sin\alpha+\sin\beta\right)^2}{\left(\sin\alpha\sin\beta\right)^2\left[1+\sin^2\alpha\sin^2\beta-\left(\sin\alpha+\sin\beta\right)^2+2\sin\alpha\left(\sin\beta\right)\right]}$

and equation is $\left(\sin x-\frac{1}{2}\right)\left(\sin x-\frac{1}{3}\right)\left(\sin x-\frac{1}{4}\right)=0$

36. B, C

Sol. Let A (x_1, y_1) be any point on $x^2 + y^2 - 40x + 399 = 0$

$\Rightarrow$ Centroid of ΔFGH is $\left(\frac{2x_1-4}{3}, 0\right)$ where $x_1 \in [19, 21]$

So, $R = \left\{(x, 0) : x \in \left[\frac{34}{3}, \frac{38}{3}\right]\right\}$

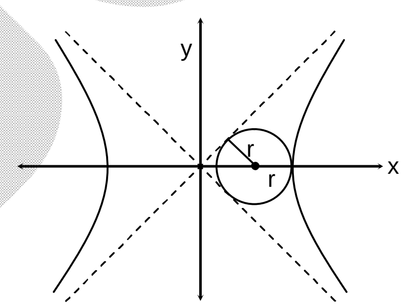
37. A, B, C

Sol. S′ is $x^2 + y^2 = 4$ excluding B and C

38. C

Sol. $r \operatorname{cosec}\theta + r = a$

$r = \dfrac{a}{\operatorname{cosec}\theta+1} = \dfrac{b}{\sqrt{1+\frac{b^2}{a^2}}+\frac{b}{a}} = e-\sqrt{e^2-1}$



39. D

Sol. $a \neq A$, $b \neq B$

40. C

Sol. If angle between the lines is θ, then $\dfrac{2x}{1-x^2}=\dfrac{3}{4}$ where $x=\sqrt{e^2-1}$ = tan θ

41. B

Sol. From the graph it is easy to observe that $x = \pi$ is the only solution

42. C

Sol. (P) $\sin^{-1}x+\cos^{-1}x \in \left[\frac{9\pi}{2}; \frac{13\pi}{2}\right]$

(Q) $\sin^{-1}x-\cos^{-1}x=\frac{\pi}{2}$

(R) $\sin^{-1}x\cdot\cos^{-1}x \in \left[5\pi^2, \frac{21}{2}\pi^2\right]$

(S) $\cos^{-1}\left(\frac{\sin^{-1}x}{5\pi}\right) \in \left[\cos^{-1}\left(\frac{7}{10}\right), \frac{7\pi}{3}\right]$

43. B

Sol. $p = \left(\sin^2 A + \sin^2 B - \sin^2 C\right)^2 \in [0, 4]$

$p' = \tan^2 F\left(\sin^2 F + \sin^2 E - \sin^2 F\right)^2 = 4\sin^2 D \sin^2 E \sin^2 F \in \left(0, \frac{27}{16}\right]$

44. B

Sol. lx + my + n + p(mx + ly + n) = 0 is a family of lines passing through fixed point $\left(\frac{-n}{l+m}, \frac{-n}{l+m}\right)$

(P) $\sqrt{\left(2-\left(\frac{-n}{l+m}\right)\right)^2 + \left(2-\left(\frac{-n}{l+m}\right)\right)^2} < 3\sqrt{2} \Rightarrow 0 < l + m - n$

(Q) Line will pass through origin so no triangle is possible

(R) Fixed point lies inside circle so no tangent can be drawn

(S) p = –1 is the only case

45. A

Sol. (P) Locus is part of hyperbola

(Q) Locus is their common tangent

(R) Locus is ellipse

(S) Locus is parabola

SECTION – B

46. 1

Sol. Domain: $x \in \left(\sqrt{2}, 2\right] \cup \{0\}$

x = 0 is not the solution

$\forall\ x \in \left(\sqrt{2}, 2\right]$, $f(x) = \sin^{-1}(x-1)^2 + \cos^{-1}\left(\frac{1}{x^2-1}\right)$ and g(x) = $\tan^{-1} x$ are strictly increasing functions, where $f\left(\sqrt{2}\right) < g\left(\sqrt{2}\right)$ and f(2) > g(2) so 1 solution

47. 4046

Sol. Let f(θ) = cos 5θ – sin θ

f(θ) = 0 ⇒ cos 5θ = sin θ = $\cos\left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow 5\theta = 2n\pi \pm \left(\frac{\pi}{2} - \theta\right)$ ..... (1)

Let g(θ) = cot 4θ – tan 2θ

g(θ) = 0 ⇒ cot 4θ = tan 2θ = $\cot\left(\frac{\pi}{2} - 2\theta\right)$

$\Rightarrow \theta = \left(\frac{2p+1}{12}\right)\pi$ ..... (2)

∀ θ ∈ [0, π], common solutions of (1) and (2) are $\frac{\pi}{12}$ and $\frac{5\pi}{12}$

⇒ Total number of solutions is 4046

48. 28

Sol. $\tan(\alpha+\beta+\gamma)=\dfrac{\sqrt{3}-\dfrac{1}{\sqrt{3}}}{1-(-3)}=\dfrac{1}{\sqrt{3}}$

or $1-3x^2=\sqrt{3}\left(3x-x^3\right)$

$\Rightarrow \dfrac{1}{\sqrt{3}}=\dfrac{3x-x^3}{1-3x^2} \Rightarrow \alpha=\dfrac{\pi}{18}, \beta=\dfrac{7\pi}{18}, \gamma=\dfrac{13\pi}{18}$

49. 27

Sol. It is an equilateral triangle

$$\Rightarrow \left(\dfrac{\dfrac{3a}{2}}{\dfrac{a}{2\sqrt{3}}}\right)^2=27$$

50. 4

Sol. Minimum value corresponds to the minimum distance between the curves $y=\sqrt{1-x^2}$ and $y=3+\sqrt{4x}$ so p = 4

51. 4

Sol. P : $y^2 = 4(x-1)$

Area = 4 sq. units